

Homework 5: Applications of Derivatives, Approximation, and Implicit Functions

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Due: Chapter 5, ETC: 1-2 hours, 100 points

Practice Problems: Work together in class

1. Suppose output is produced according to

$$Y = F(K) = 12K^{1/2}.$$

The current level of capital is $K_0 = 100$.

- (a) Find $F'(K)$.
- (b) Find $F'(100)$.
- (c) Use the linear approximation

$$F(K) \approx F(100) + F'(100)(K - 100)$$

to approximate $F(104)$.

- (d) Compute the exact value $F(104)$.
- (e) Is the approximation close to the exact value?

2. Suppose demand for a good is given by

$$Q(p) = 100 - 5p^{1/2}.$$

The current price is $p_0 = 16$.

- (a) Find $Q'(p)$.
- (b) Find $Q'(16)$.
- (c) Use a linear approximation to estimate the change in demand when price rises from 16 to 17.
- (d) Approximate $Q(17)$ using your answer from part (c).
- (e) Interpret the sign of $Q'(16)$ economically.

3. Suppose utility from income is

$$u(m) = \ln m.$$

The current income level is $m_0 = 100$.

- (a) Find $u'(m)$.
- (b) Use a linear approximation around $m_0 = 100$ to approximate $u(105)$.
- (c) Compute the exact value $\ln(105)$ using a calculator.
- (d) Is the approximation larger or smaller than the exact value?
- (e) Why might this happen given the shape of the log function?

4. Consider the implicit equation

$$xy = 20.$$

Suppose this equation implicitly defines y as a function of x .

- (a) Use implicit differentiation to find $\frac{dy}{dx}$.
- (b) Solve explicitly for y as a function of x .
- (c) Differentiate the explicit function and compare your answer with part (a).
- (d) Find $\frac{dy}{dx}$ at the point $(x, y) = (4, 5)$.
- (e) If $xy = 20$ is an indifference curve, interpret $\frac{dy}{dx}$ at $(4, 5)$.

5. Suppose a consumer's indifference curve is described implicitly by

$$U(x, y) = x^{1/2}y^{1/2} = 10.$$

- (a) Compute U_x and U_y .
- (b) Use the implicit function theorem formula

$$\frac{dy}{dx} = -\frac{U_x}{U_y}$$

to find the slope of the indifference curve.

- (c) Simplify your answer.
- (d) Evaluate the slope at $(x, y) = (25, 4)$.
- (e) Interpret the sign of the slope economically.

Homework Problems: Submit individually

6. Suppose a firm's cost function is

$$C(q) = 50 + 10q + q^2.$$

The firm currently produces $q_0 = 20$ units.

- (a) Find the marginal cost function $C'(q)$.
- (b) Find $C'(20)$.
- (c) Use a linear approximation to estimate the increase in cost when output rises from 20 to 21.
- (d) Compute the exact increase in cost:

$$C(21) - C(20).$$

- (e) Explain why marginal cost gives an approximation to the change in total cost.

7. Suppose output is produced according to

$$Y = F(L) = 30L^{1/3}.$$

The current level of labor is $L_0 = 27$.

- (a) Find $F'(L)$ and $F''(L)$.
- (b) Use the linear approximation around $L_0 = 27$ to approximate $F(28)$.
- (c) Use the quadratic approximation

$$F(L) \approx F(27) + F'(27)(L - 27) + \frac{1}{2}F''(27)(L - 27)^2$$

to approximate $F(28)$.

- (d) Compute the exact value $F(28)$ using a calculator.
- (e) Which approximation is closer?

8. Suppose a firm faces price p and per-unit tax t . Its profit is

$$\pi(q, t) = pq - C(q) - tq,$$

where

$$C(q) = q^2.$$

- (a) Find the first-order condition for profit maximization.
- (b) Solve for the firm's optimal output $q^*(t)$ as a function of p and t .
- (c) Find $\frac{dq^*}{dt}$.
- (d) Suppose $p = 40$ and $t = 4$. Find $q^*(4)$.
- (e) Use linear approximation to estimate the new optimal output if the tax rises from 4 to 5.
- (f) Interpret the sign of $\frac{dq^*}{dt}$.

9. Suppose equilibrium in a market is described by the equation

$$D(p, a) = S(p),$$

where demand is

$$D(p, a) = a - 2p$$

and supply is

$$S(p) = 10 + p.$$

Here a is a demand-shift parameter.

- (a) Rewrite the equilibrium condition as

$$F(p, a) = 0.$$

- (b) Solve explicitly for the equilibrium price $p^*(a)$.

- (c) Find $\frac{dp^*}{da}$ using your explicit solution.

- (d) Now use the implicit function theorem:

$$\frac{dp^*}{da} = -\frac{F_a}{F_p}.$$

Verify that you get the same answer.

- (e) Interpret $\frac{dp^*}{da}$ economically.

10. **Slightly harder problem.** Suppose a consumer's indifference curve is given implicitly by

$$F(x, y) = x^{1/2} + y^{1/2} - 10 = 0.$$

- (a) Compute F_x and F_y .

- (b) Use the implicit function theorem to find

$$\frac{dy}{dx}.$$

- (c) Evaluate $\frac{dy}{dx}$ at the point $(x, y) = (16, 36)$.

- (d) Use a linear approximation to estimate how much y must change if x increases from 16 to 17, while staying on the same indifference curve.

- (e) Check whether your answer makes economic sense: should y rise or fall when x rises along an indifference curve?

- (f) Compute the exact value of y when $x = 17$ by solving

$$17^{1/2} + y^{1/2} = 10.$$

Compare the exact change in y to the approximate change from part (d).